

# LECTURE-24 TRIGONOMETRIC FOURIER SERIES



# Symmetry in Trigonometric Fourier Series

## Even & Odd symmetry

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin n\omega_0 t dt$$

if  $x(t) = \text{even}$   $x(t) \sin n\omega_0 t : \text{odd } \forall n$

$$b_n = 0 \quad a_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad a_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos n\omega_0 t dt$$

if  $x(t) = \text{odd}$   $x(t) \sin n\omega_0 t = \text{even}$

$x(t) \cos n\omega_0 t = \text{odd}$

$$a_0 = 0 \quad a_n = 0$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin n\omega_0 t dt$$

Fourier series of even function  
only contains dc & cosine &  
for odd functions only  
sine terms.

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# Functions with rotational symmetry

A function  $x(t)$  having time period  $T$  is said to have rotational symmetry or half wave symmetry if

$$x\left(t + \frac{T}{2}\right) = -x(t)$$

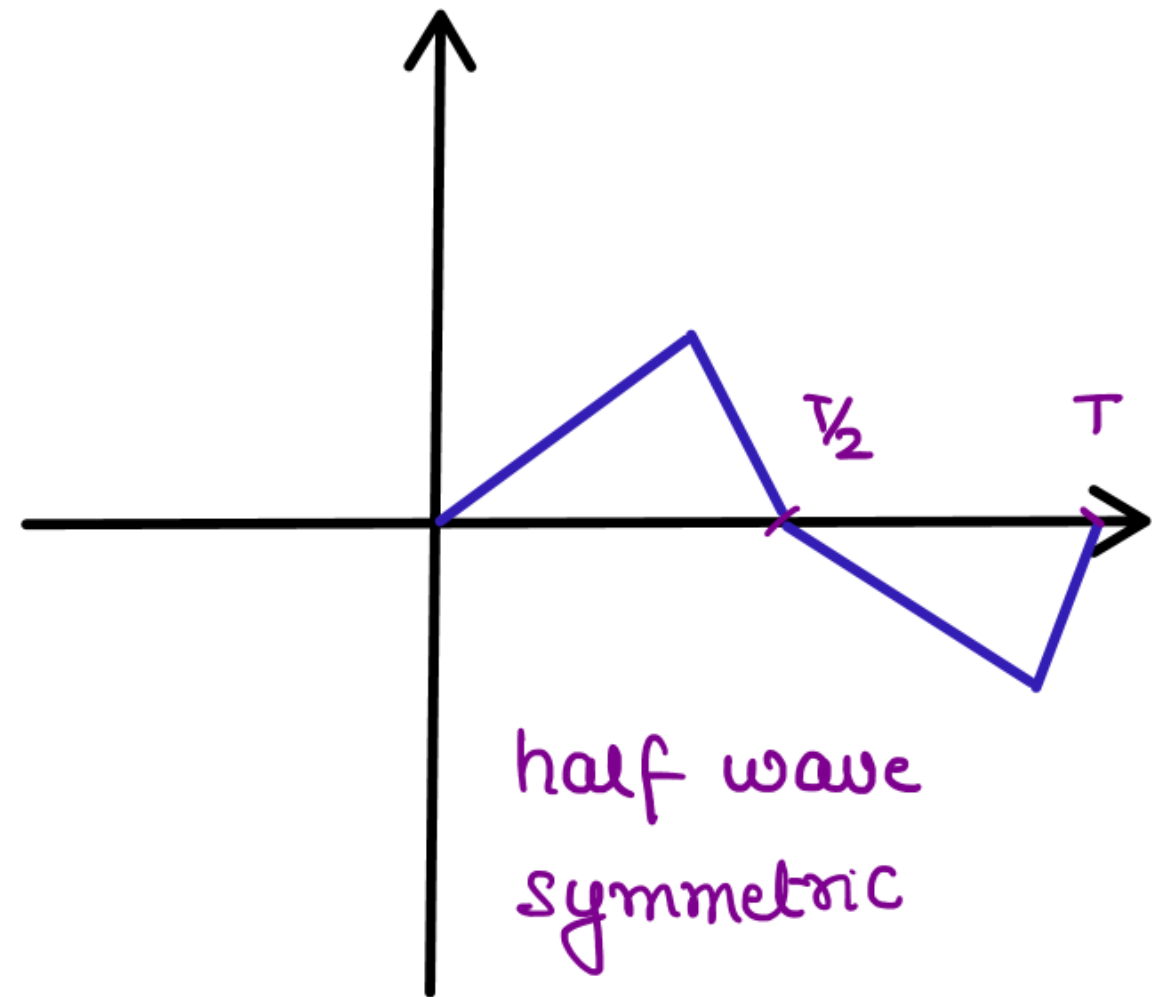
Waveform of signal from  $T/2$  upto  $T$  should be -ve of waveform from 0 to  $T/2$ .

In such a case only odd harmonics exist

$$a_n = b_n = 0 \quad \forall \quad n = \text{even} \quad a_0 = 0$$

$$a_n, b_n \neq 0 \quad \forall \quad n = \text{odd}$$

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# Hidden half wave symmetry

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for signals with non-zero average value or dc component

Step-1: Compute average value using waveform

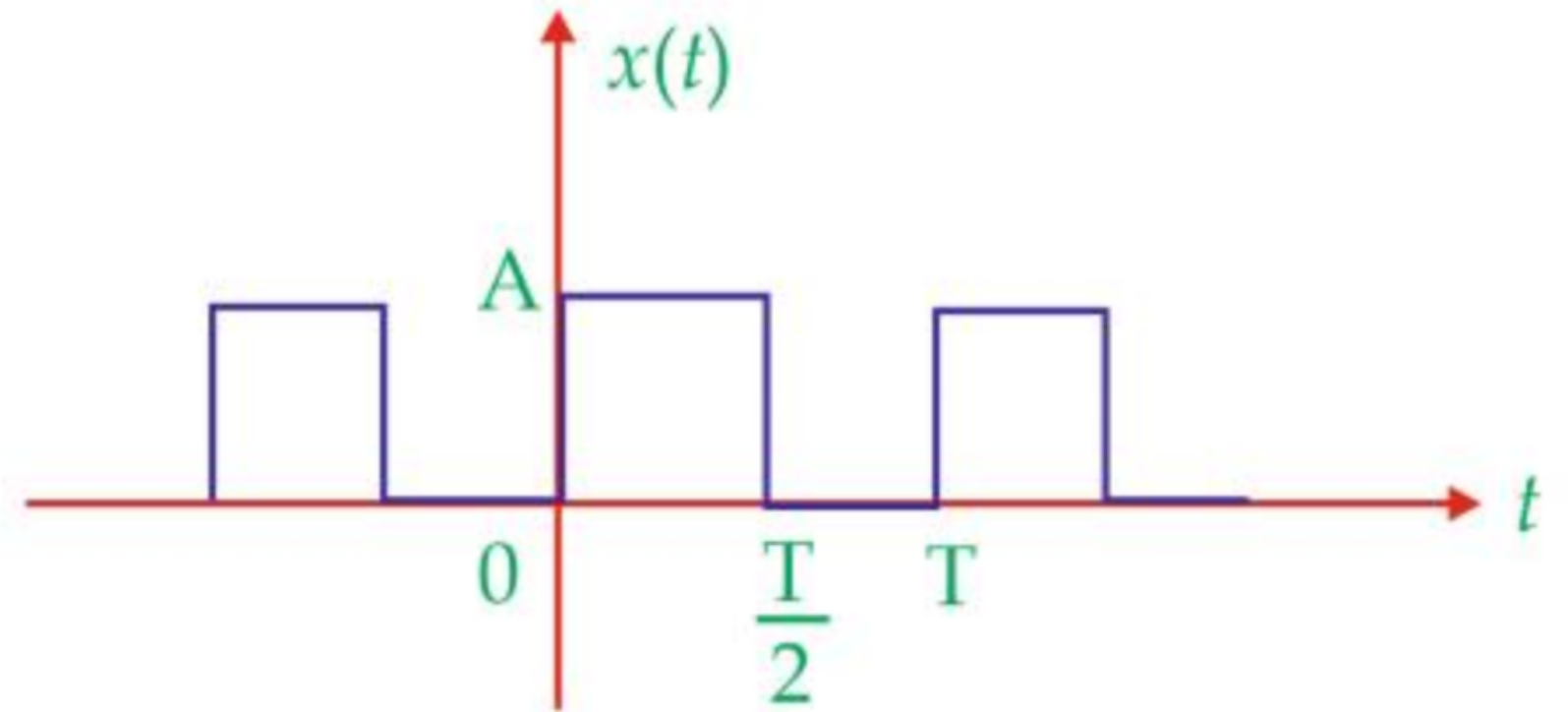
Step-2: Subtract average value from  $x(t)$  & shift waveform vertically

Step-3: Check for HWS



average  $\neq 0$

$$\begin{aligned}\text{average} &= \frac{1}{T} \int_0^T x(t) dt \\ &= \frac{1}{T} \int_0^{T/2} A dt = \frac{A \cdot T/2}{T} \\ &= A/2\end{aligned}$$



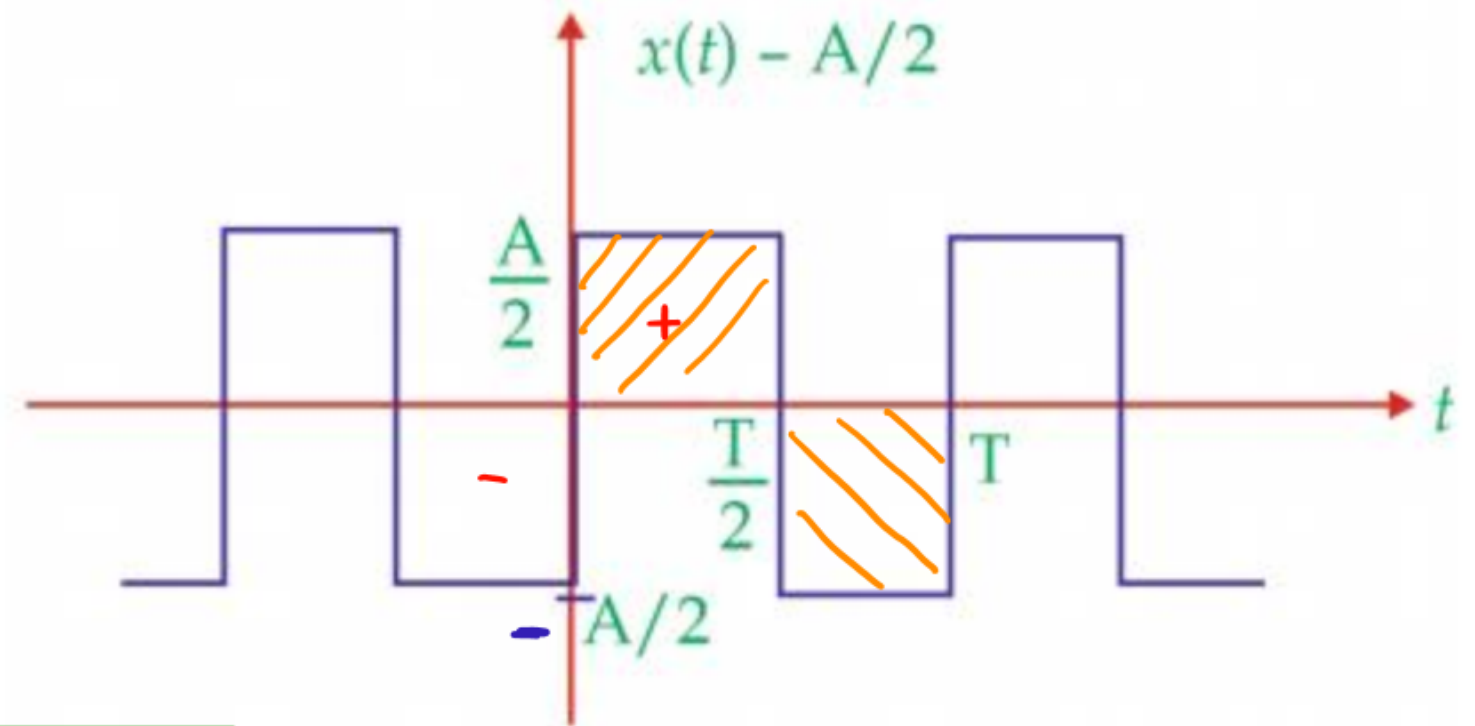
By subtracting  $A/2$  signal shifts downwards by  $A/2$

$$x(t) - A/2 : \text{HWS}$$

contains only odd harmonic

Note: Shifting a signal vertically upwards or downwards does not affect harmonics but only affects dc value.

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odd  
contains only sine  
terms of odd  
order.  
& dc component =  $A/2$

## Question-02

Describe Fourier series coefficient for following signals

(a)  $x(t)$  : even & half wave symmetric  $\rightarrow$  even: only cosine terms

(b)  $x(t)$  : ~~odd~~ & half wave symmetric HWS: odd harmonics

$\downarrow$

odd: only sine terms

HWS: odd harmonics

$x(t)$  :  $b_n \quad \forall n = \text{odd}$

$$a_0 = a_n = 0$$

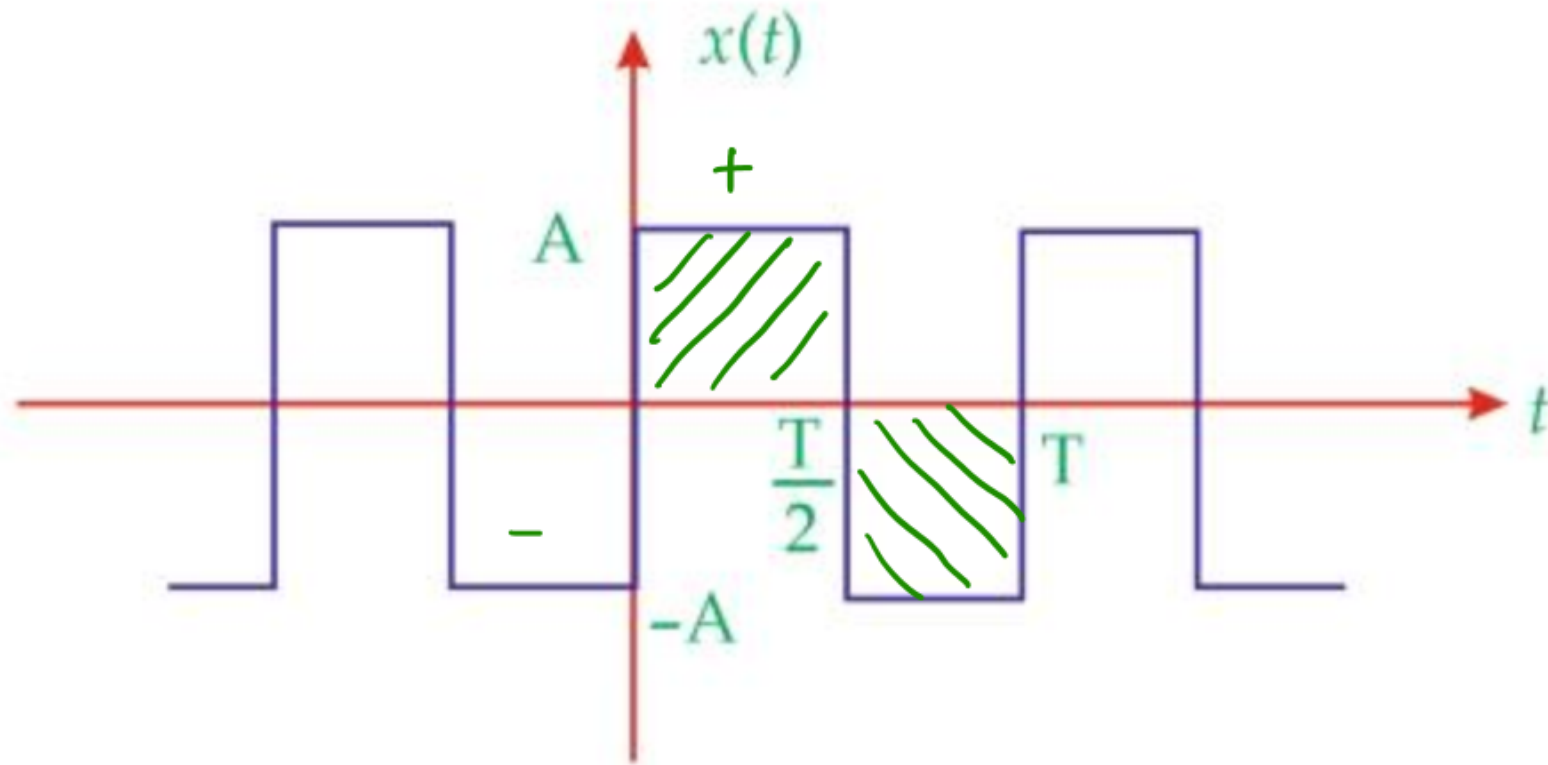
$x(t)$  :  $a_0, a_n \quad \forall n = \text{odd}$

$$b_n = 0$$



### Question-03

Describe Fourier series coefficient for following signals



half wave symmetric  $\rightarrow$  odd harmonics

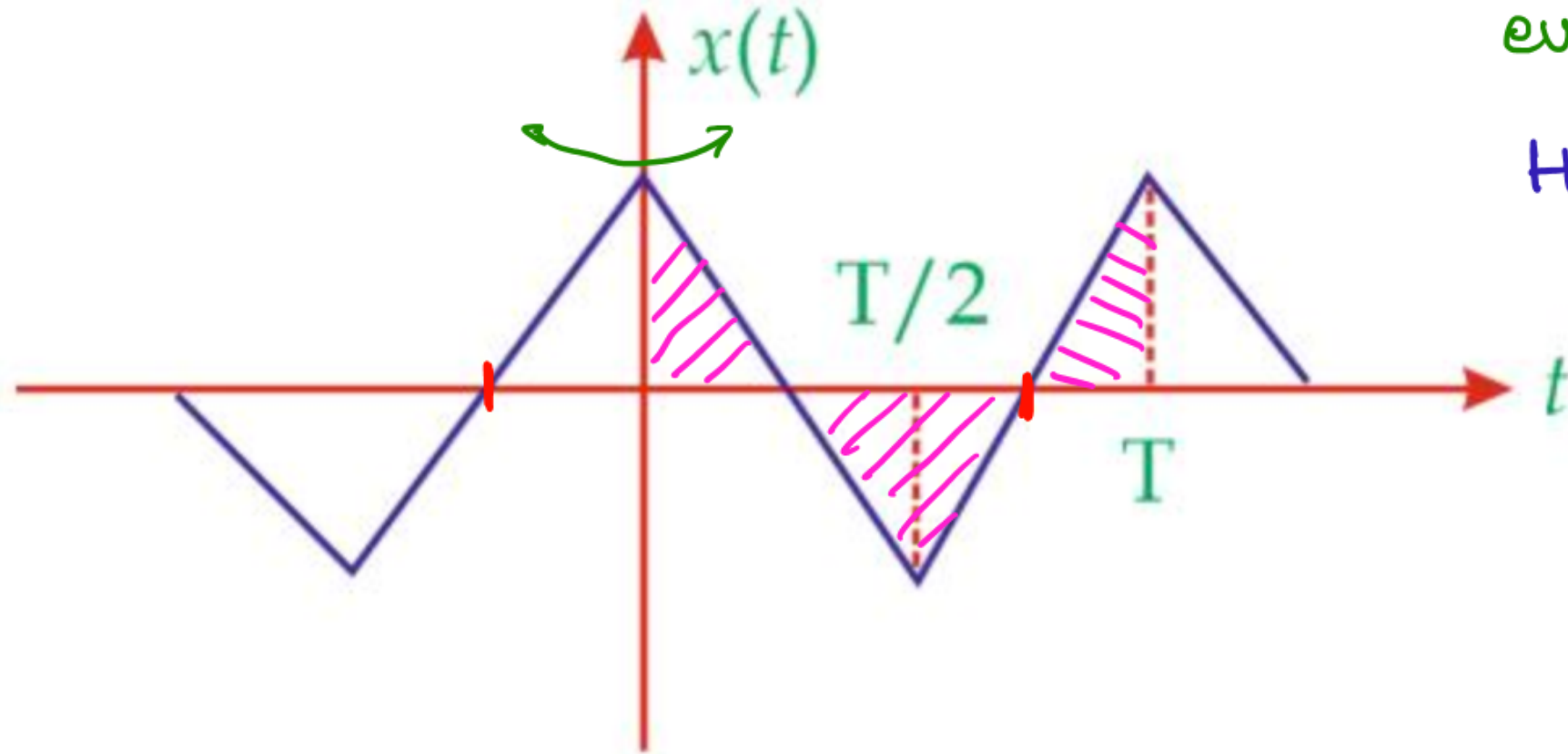
odd  $\rightarrow$  only sine terms

contains only sine terms of  
odd order

$b_n, \forall n = \text{odd}$

## Question-04

Describe Fourier series coefficient for following signals



area = 0

$$\text{avg} = \frac{1}{T} \int_{\langle T \rangle} x(t) dt = \frac{\text{area}}{\text{period}}$$

even : dc + cosine terms

HWS : odd harmonics

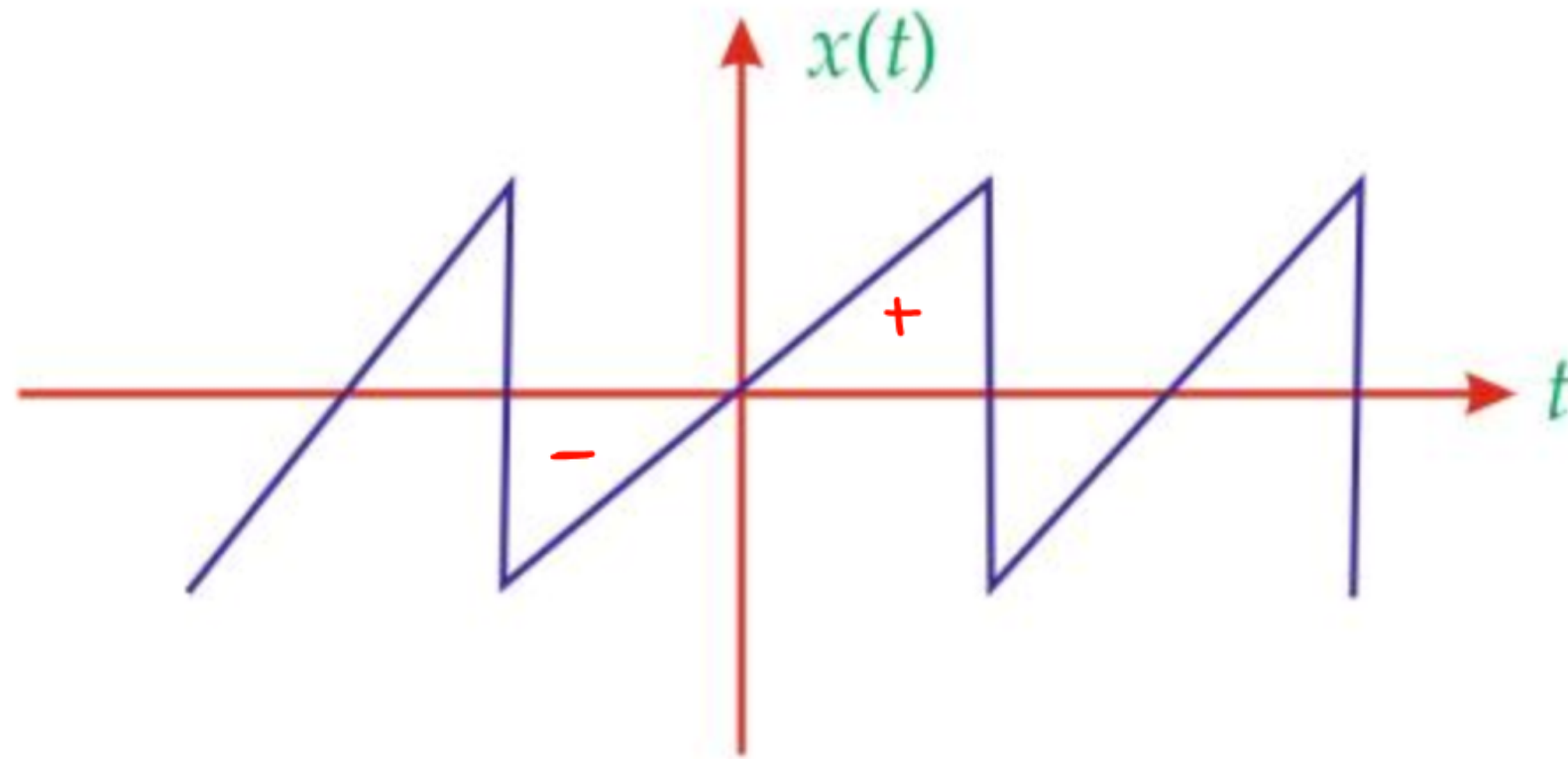
$$dc = 0$$

only cosine terms of odd order

$$a_n, \forall n = \text{odd}$$

## Question-05

Describe Fourier series coefficient for following signals



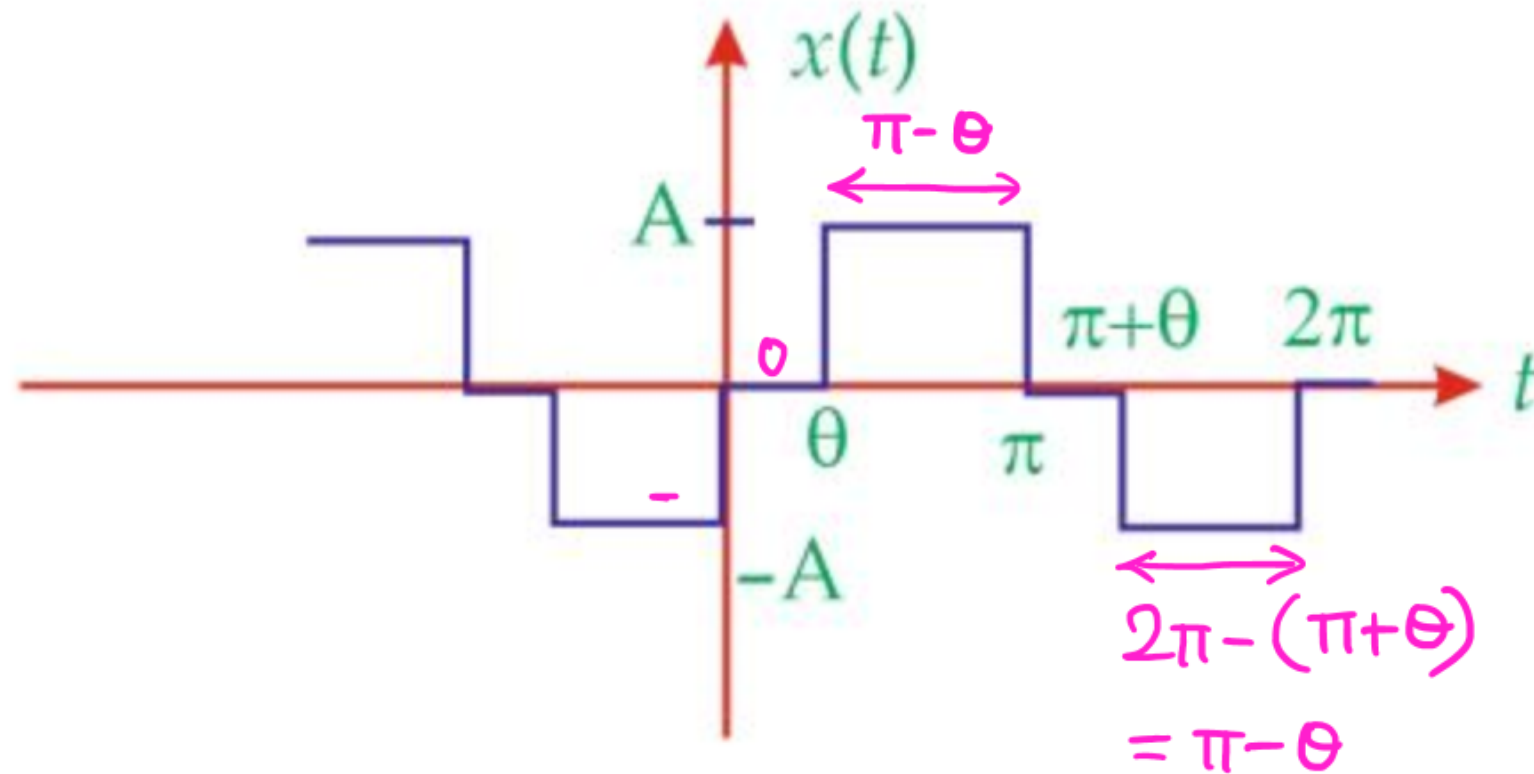
odd : only sine terms

not HWS

$b_n \neq 0$

## Question-06

Describe Fourier series coefficient for following signals



neither odd nor even

$$a_n, b_n \neq 0$$

$$\text{area} = 0$$

$$\text{avg} = 0$$

$$a_0 = 0$$

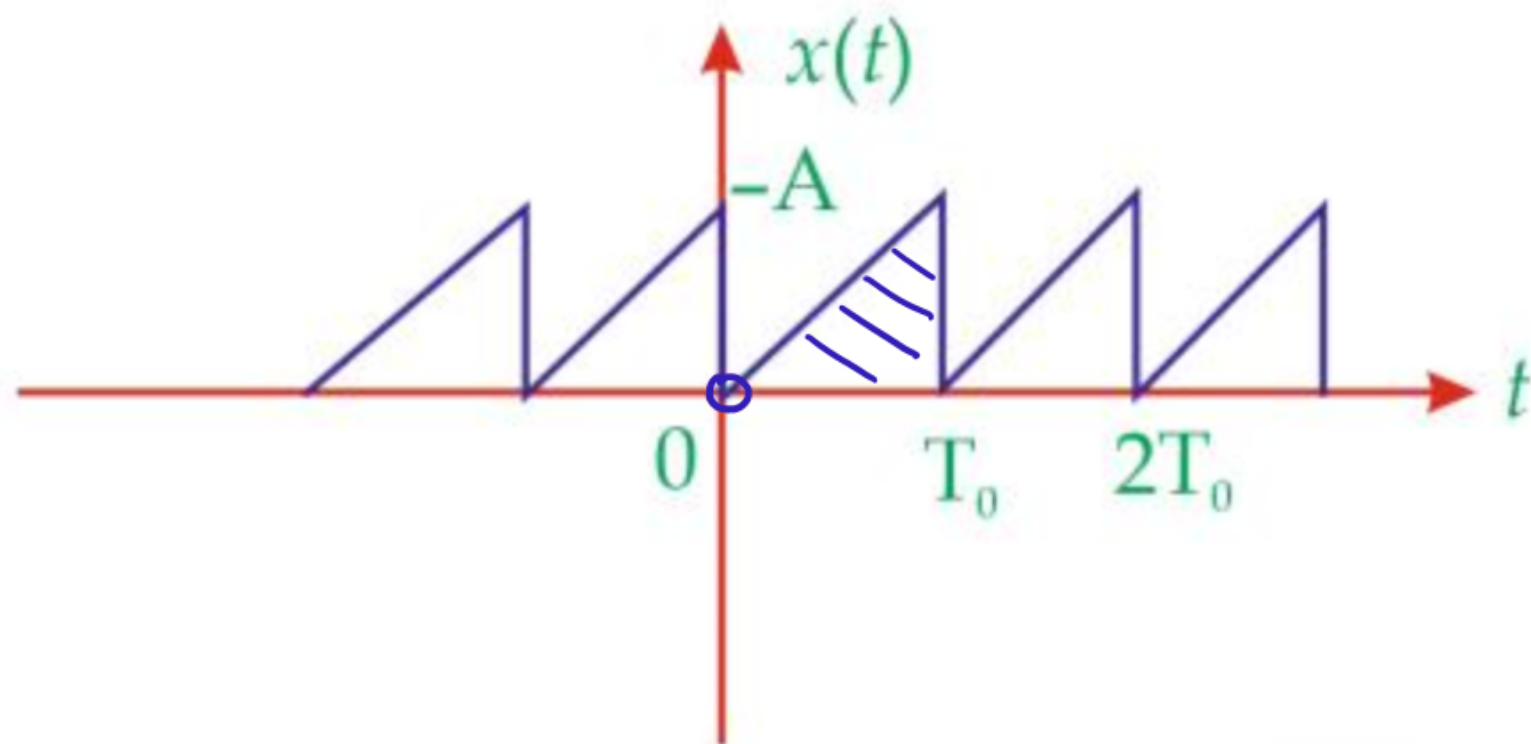
half wave symmetric  
only odd harmonics

$$a_n, b_n \neq 0 \text{ for } n = \text{odd}$$



## Question-07

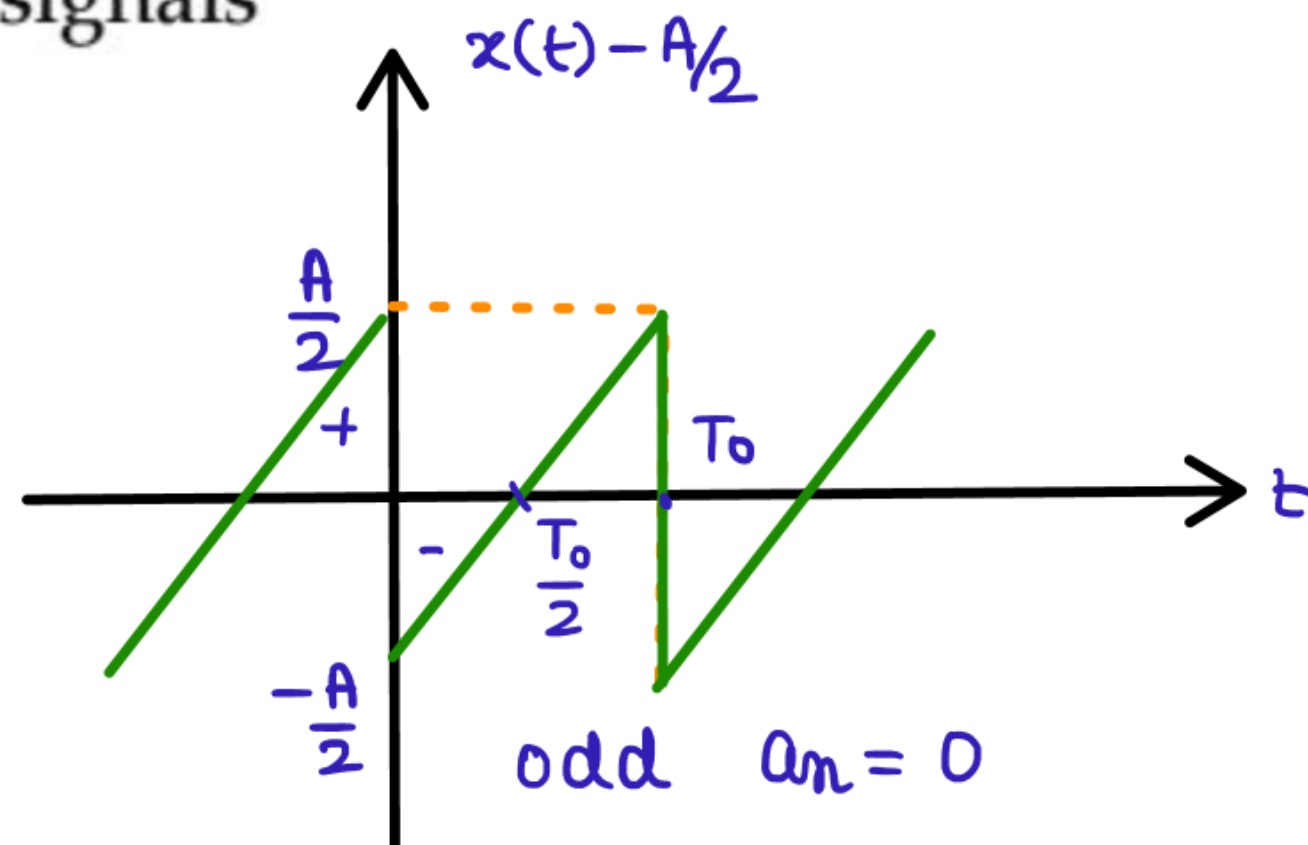
Describe Fourier series coefficient for following signals



Area  $\neq 0$       avg  $\neq 0$

$a_0 \neq 0$

$$\text{avg} = \frac{\text{area}}{\text{period}} = \frac{1}{2} \frac{A \times T_0}{T_0} = A/2$$



odd  $a_n = 0$

not HWS

dc + sine terms  $\forall n$

## Question-08

Check if trigonometric Fourier series exists for signal below:

$$x(t) = 4\cos(\pi/2 t) - 5\sin(2t + 30^\circ)$$

$$2\pi/\pi/2 = 4$$

$$2\pi/2 = \pi$$

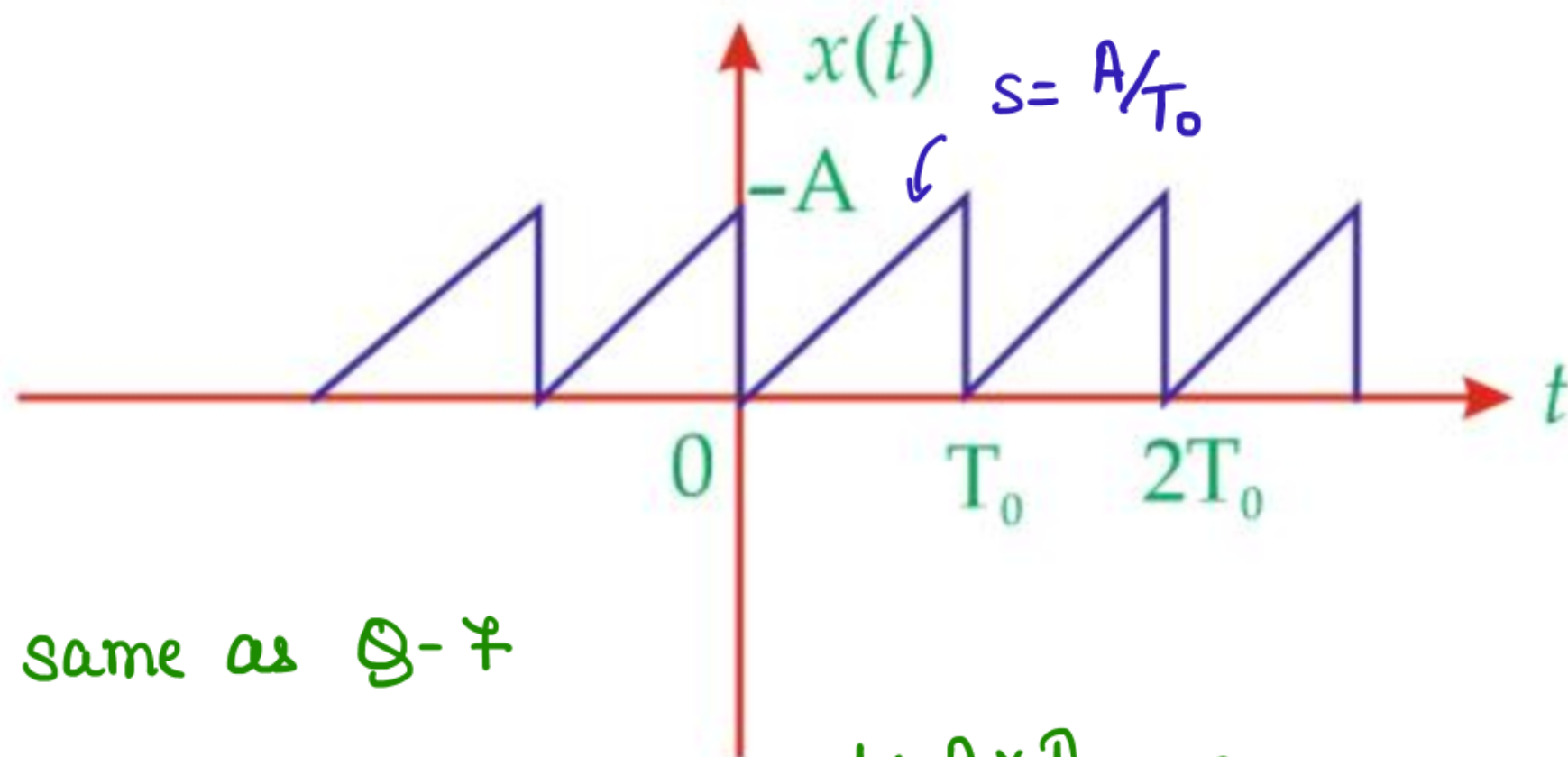
$$T_1/T_2 = 4/\pi : \text{irrational}$$

$x(t)$  : aperiodic

FS does not exist

## Question-09

Find trigonometric Fourier series for signal shown:



same as Q-7

$$\text{Average} = \frac{\text{area}}{\text{period}} = \frac{1}{2} \frac{A \times T_0}{T_0} = A/2$$

$$a_0 = A/2$$

$x(t) - A/2$ : odd only  $b_n$  exists

$$x(t) = \frac{At}{T_0} \quad 0 < t < T_0$$

$$b_n = \frac{2}{T_0} \int_0^{T_0} x(t) \sin n\omega_0 t \, dt$$

$$= \frac{2}{T_0} \int_0^{T_0} \frac{At}{T_0} \sin n\omega_0 t \, dt$$

$$= \frac{2A}{T_0} \int_0^{T_0} \underset{\substack{\uparrow \\ I}}{t} \cdot \underset{\substack{\downarrow \\ II}}{\sin n\omega_0 t} \, dt$$

$$b_n = \frac{2A}{T_0^2} \left[ t \left( -\frac{\cos n\omega_0 t}{n\omega_0} \right) - \left( -\frac{\sin n\omega_0 t}{(n\omega_0)^2} \right) \right]_0^{T_0}$$

$$= \frac{2A}{T_0^2} \left[ T_0 \left( -\frac{\cos n\omega_0 T_0}{n\omega_0} \right) + \frac{\sin n\omega_0 T_0}{(n\omega_0)^2} - 0 - 0 \right]$$

$$\omega_0 = 2\pi/T_0 \quad \omega_0 T_0 = 2\pi$$

$$b_n = \frac{2A}{T_0^2} \left( T_0 \left( -\frac{\cos 2n\pi}{n\omega_0} \right) + \frac{\sin 2n\pi}{(n\omega_0)^2} \right)$$

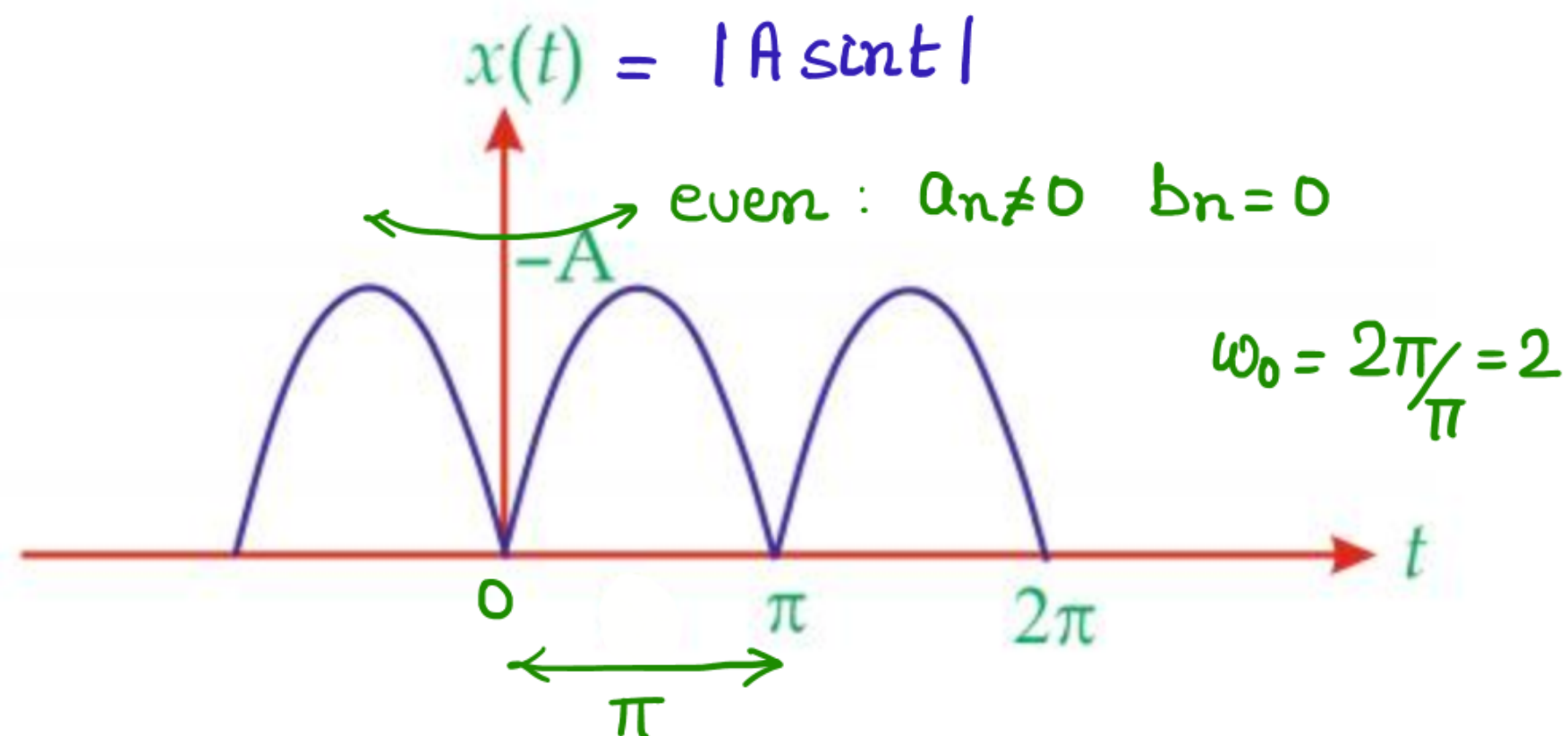
$$= -\frac{2A}{T_0} \times \frac{1}{n\omega_0} = \frac{-2A}{2n\pi} = -A/n\pi$$

$$x(t) = A/2 - \sum_{n=1}^{\infty} A/n\pi \sin n\omega_0 t$$



## Question-10

Find trigonometric Fourier series for signal shown:



area  $\neq 0$

avg  $\neq 0$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} A \sin t \, dt = \frac{A}{\pi} (-\cos t)_0^{\pi} = 2A/\pi$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} A \sin t \cdot \cos n\omega_0 t \, dt \\ &= \frac{2A}{\pi} \int_0^{\pi} \frac{1}{2} [\sin(2n+1)t - \sin(2n-1)t] \, dt \\ &= \frac{A}{\pi} \left[ -\frac{\cos(2n+1)t}{(2n+1)} + \frac{\cos(2n-1)t}{(2n-1)} \right]_0^{\pi} \end{aligned}$$

$$a_n = \frac{A}{\pi} \left[ \frac{-\cos(2n+1)\pi}{(2n+1)} + \frac{\cos(2n-1)\pi}{(2n-1)} + \frac{1}{(2n+1)} - \frac{1}{(2n-1)} \right]$$

$$= \frac{A}{\pi} \left[ \frac{1}{(2n+1)} - \frac{1}{(2n-1)} + \frac{1}{(2n+1)} - \frac{1}{(2n-1)} \right]$$

$$= 2 \frac{A}{\pi} \left[ \frac{1}{(2n+1)} - \frac{1}{(2n-1)} \right]$$

$$= 2 \frac{A}{\pi} \left[ \frac{-2}{4n^2-1} \right]$$

$$= \frac{-4A}{\pi(4n^2-1)}$$

$$x(t) = \frac{2A}{\pi} - \sum_{n=1}^{\infty} \frac{4A}{\pi(4n^2-1)} \cos n\omega_0 t$$

## Question-11

Compute trigonometric Fourier series coefficients for following signals.

$$x(t) = 2\sin\left(\frac{\pi}{4}t\right) + 4\cos\left(\frac{\pi}{5}t\right) \quad \leftarrow \quad \omega_1 = \pi/4 \quad \omega_2 = \pi/5 \quad \omega_0 = \text{HCF}(\omega_1, \omega_2) \\ = \frac{\text{HCF}(\pi, \pi)}{\text{LCM}(4, 5)} = \pi/20$$

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$$x(t) = 2 - 3\sin(7t) + 5\cos\left(\frac{3t}{7} - 30^\circ\right) - 10\cos(0.2t + 45^\circ) \quad \dots\dots\dots$$

$\hookrightarrow 1/5$

$$\omega_1 = 5\omega_0 : 5^{\text{th}} \text{ har}$$

$$\omega_2 = 4\omega_0 : 4^{\text{th}} \text{ harmonic}$$

$$b_5 = 2$$

$$a_4 = 4$$

$$\omega_0 = \text{HCF}\left(7, 3/7, 1/5\right) = \frac{\text{HCF}(7, 3, 1)}{\text{LCM}(7, 5)} = 1/35$$

$$7 = 245\omega_0 \quad 3/7 = 15\omega_0 \quad 1/5 = 7\omega_0$$

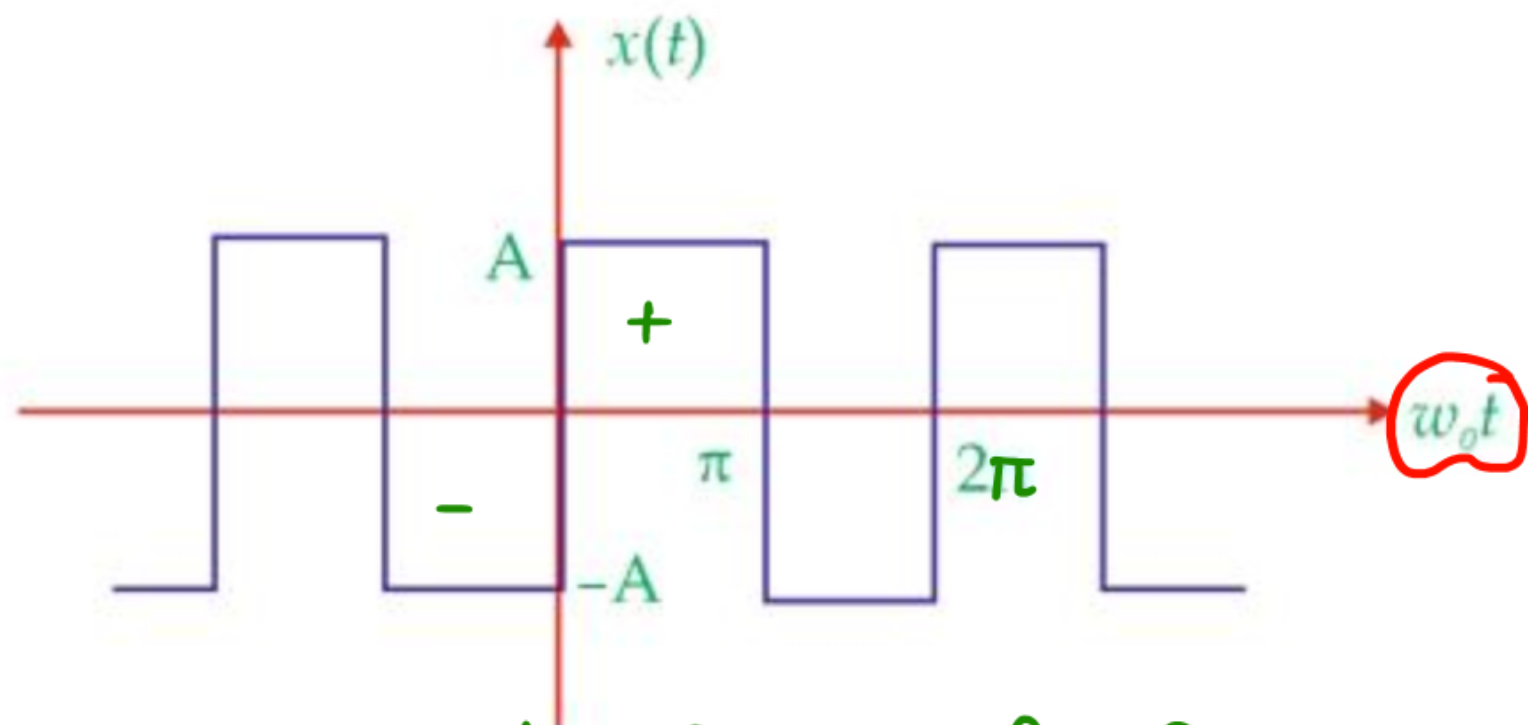
$$\tau_0 = 2$$

$$\tau_{245} = b_{245} = -3$$

$$\tau_{15} = 5 \quad \tau_7 = -10$$

## Question-12

Compute trigonometric Fourier series follow



odd:  $a_n = 0$   $a_0 = 0$

HWS : only odd harmonics

$$b_n = \frac{2}{2\pi} \int_0^{2\pi} x(t) \sin n\omega_0 t \, d(\omega_0 t)$$

$$= \frac{2}{2\pi} \left[ \int_0^{\pi} A \sin n\omega_0 t \, d(\omega_0 t) - \int_{\pi}^{2\pi} A \sin n\omega_0 t \, d(\omega_0 t) \right]$$

$$= \frac{A}{\pi} \left[ \left( \frac{-\cos n\omega_0 t}{n} \right)_0^{\pi} - \left( \frac{-\cos n\omega_0 t}{n} \right)_{\pi}^{2\pi} \right]$$

$$b_n = \frac{A}{n\pi} \left[ 1 - \cos n\pi + \cos 2n\pi - \cos n\pi \right]$$



$$b_n = \frac{2A}{n\pi} (1 - \cos n\pi)$$

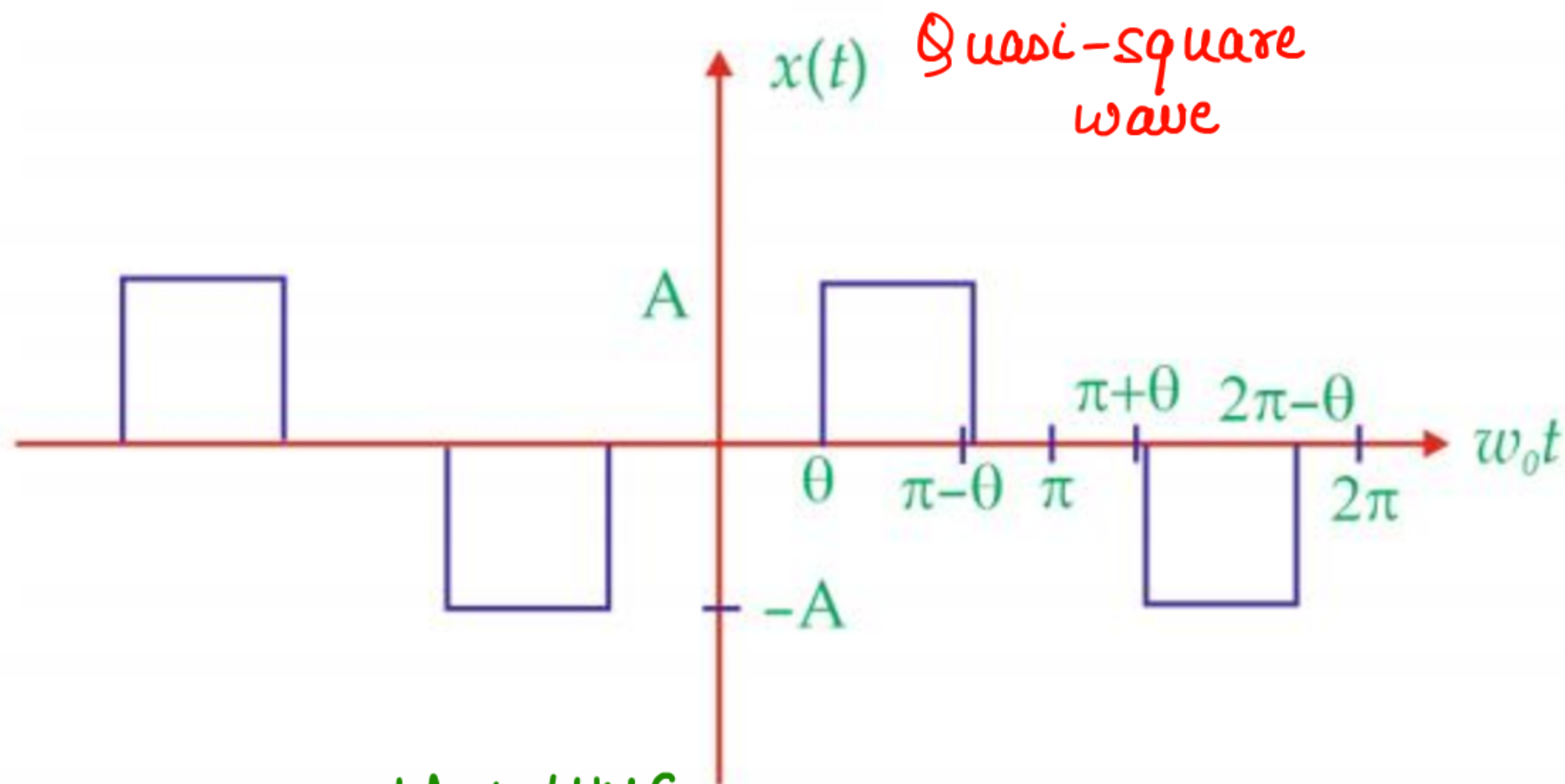
$$n = \text{even} \quad \cos n\pi = 1 \quad b_n = 0$$

$$n = \text{odd} \quad \cos n\pi = -1 \quad b_n = \frac{4A}{n\pi}$$

$$x(t) = \sum_{n=1,3,5}^{\infty} \frac{4A}{n\pi} \sin n\omega_0 t$$

## Question-13

Compute trigonometric Fourier series coefficients for following signals.



odd + HWS  
only sine terms of  
odd order

$$b_n = \frac{2}{2\pi} \int_0^{2\pi} x(t) \sin n\omega_0 t d(\omega_0 t)$$

$$= \frac{2}{2\pi} \left[ \int_{\theta}^{\pi-\theta} A \sin n\omega_0 t d(\omega_0 t) - \int_{\pi+\theta}^{2\pi-\theta} A \sin n\omega_0 t d(\omega_0 t) \right]$$

$$b_n = \frac{A}{\pi} \left[ \left( -\frac{\cos n\omega_0 t}{n} \right)_{\theta}^{\pi-\theta} - \left( -\frac{\cos n\omega_0 t}{n} \right)_{\pi+\theta}^{2\pi-\theta} \right]$$

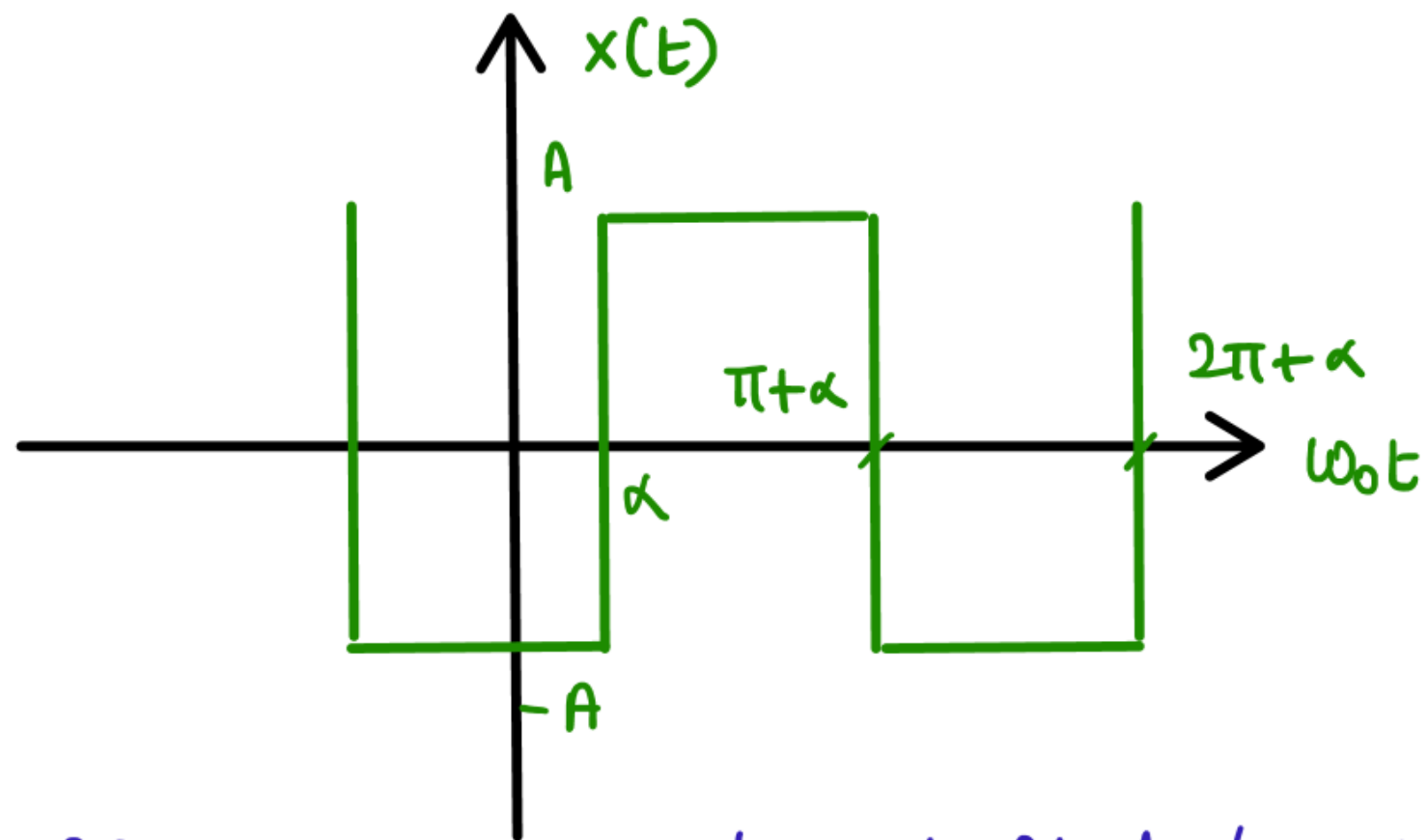
$$= \frac{A}{n\pi} \left[ \cos n\theta - \cos(n\pi - n\theta) + \cos(2n\pi - n\theta) - \cos(n\pi + n\theta) \right]$$

$$n = \text{odd} \quad \cos(2n\pi - n\theta) = \cos n\theta$$

$$\cos(n\pi \pm n\theta) = -\cos n\theta$$

$$b_n = \frac{4A}{n\pi} \cos n\theta$$

$$x(t) = \sum_{n=1,3,5}^{\infty} \frac{4A}{n\pi} \cos n\theta \sin n\omega_0 t$$



square wave right shifted by  $\omega_0 t = \alpha$

square wave : 
$$\sum_{n=1,3,5}^{\infty} \frac{4A}{n\pi} \sin n\omega_0 t$$

right shift : 
$$\sum_{n=1,3,5}^{\infty} \frac{4A}{n\pi} \sin [n(\omega_0 t - \alpha)]$$

$$= \sum_{n=1,3,5}^{\infty} \frac{4A}{n\pi} \sin (n\omega_0 t - n\alpha)$$

← polar form